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Hybrid functional materials forming the metal structure with optimal imperfection for storage of hydrogen in hydride form ☆

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ABSTRACT

The paper investigates the samples of Ni–B and Ni–In composites in the form of films synthesized by an electrochemical method having a 3high degree of defectiveness and shows the electrochemical processes for their preparation, in which the experimental data of the influence doping additions of boron and indium on the hydrogen permeability of synthesized electrochemical complexes based on nickel Ni_x-B_y-$H_z$and Ni_x-In_y-H_z are given. It is shown that by creating traps (structural, impurity) for hydrogen by introducing additional elements into the metal structure or by changing the structure (intermetallides), also by other methods, the hydrogen solubility of the metal can be changed to a greater or lesser extent depending on the technical requirements. Ni–B samples were obtained in the sulfamate nickel electrolyte using boron compounds of the class of higher polyhedral borates $Na_2B_{10}H_{10}$. The use of nanoforming boron additives provides measured hydrogen content in Ni_x-B_y-$H_{z\ samples}$ about of 600 cm^3/100 g. Nickel-indium composites were obtained by electrolytic deposition on copper substrates (0,05 mm thick), with electrolyte composition: $NiSO_4 \times 7H_2O = 140$ g/L; $Na_2SO_4 \times 10H_2O = 20$ g/L; and $In_2(SO_4)_3$ which concentration varied from 1 g/L to 12 g/L. Variations in the amount of $In_2(SO_4)_3$ ensured the production of Ni–In composites with different component ratios. The obtained data made it possible to select the optimal mode of deposition of Ni–In plating in the $In_2(SO_4)_3$ electrolyte. In order to determine the effect of each of the electrodepositing parameters on the properties of the samples, only one parameter varies: either the cathode current density or the concentration of indium in the electrolyte. The samples of different concentrations of In are synthesized and studied. Their phase composition was determined. X-Ray phase analysis reveals in the Ni–In composites synthesized from electrolytes with $In_2(SO_4)_3$ concentration of more than 2 g/L, a phase corresponding to the intermetallide η-$In_{27}Ni_{10}$.

For the first time in the practice of studying sorption, electrochemical composites Ni_x-In_y, including subsequent thermal desorption of hydrogen, are studied by implanting deuterium into the samples. It is demonstrated spectra of deuterium thermal desorption from Ni_{70}-$In_{30}D_x$ composites, which allowed determining the temperature ranges of desorption of ion-implanted deuterium depending on the dose of implantation. It is confirmed that a $Ni_{70}In_{30}$ composite capable of retaining doped deuterium (hydrogen) has been obtained. It is shown that it is permissible to obtain samples of a Ni_x-In_y-D_z composite with a deuterium content of up to 2 at. D/at. Met., that corresponds to 5,3 wt% (for composites of this composition).

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Nomenclature

Greek letters

η	Overvoltage of electrodepositing of an element on an electrode, V.
λ	The fraction of the current carried by one of the ions (transfer number), A
χ	Metal compressibility factor
E	The 5-th phase corresponding to the inns intermetallide InNi
Γ	The third phase corresponding to the intermetallic compounds of composition $InNi_3$
δ'	The 4-th phase corresponding to intermetallide In_3Ni_2

Latin letters

A	Characteristic of the degree of irreversibility of the process on the electrode, V.
B	Characteristics of the electrochemical process, V.
F	Faraday constant, equal to 96,485 Kl
I	Current density, A/cm^2
R	Universal gas constant, equal to 8,31 J/mol•K
V	Volt-unit of measurement of overvoltage in Volts
g/L	Concentration of the substance, gram in 1 L of solution
x	Content of elements in mass percentages
y	Content of elements in mass percentages
z	Content of elements in mass percentages
kJ	Kilojoules-unit of measurement of the thermal effect of the reaction
Å	Is a unit of measurement, the Angstrom (1 Å = 10^{-10} m)

Lower indices

K	Cathode
el-te	Electrolyte
BP5/20	Tungsten-rhenium thermocouple for high temperature measurement (95% W, 5% Re)
Ni_x-B_y-H_z	Nickel-boron-hydrogen
Ni_x-In_y-H_z	Nickel-indium-hydrogen
at. D/at. Met.	The ratio of the deuterium atom to the metal atom
wt.%	Is the % (mass.) or mass percent
Ppm	Is a unit less unit
X- Ray	X-Ray radiation
cm^3/100 g	Content of hydrogen in cubic centimeters per 100 g of sample
A/cm^2	Current density in amperes per square centimeter
mA/cm^2	Unit of measurement of current density Milliampere (mA per 1 square centimeter)
t_{el-ta} °C	The temperature of the electrolyte in degrees Celsius
10^{-6} mm	Micrometer-a longitudinal unit of length in the International system of units (SI). Equal to one millionth of a meter (10^{-6} m)
5 × 45 × 0,1 mm^3	Is the size of the sample in cubic millimeters
12 keV	Energy Kev in kiloelectrovolts (1 eV = 1602•10^{-19} j)
dm^2	Area in square decimeters
8 μm	The thickness of the composite layer, 8 μm (Micrometer-a longitudinal unit of length in the International system of units (SI). Equal to one millionth of a meter (10^{-6} m or 10^{-3} mm) 1 μm = 0,001 mm = 0,0001 cm = 0.000001 m)
η-$In_{27}Ni_{10}$	Phase corresponding to the composition of intermetallides $In_{27}Ni_{10}$

Abbreviations

SHF	Static hydrogen fatigue
BCC	Body–centered cubic crystal lattice.
FCC	Face cubic crystal lattice.
TDS	Thermal Desorption Mass Spectrometry
DRON-2,0	The unit is configured to work with copper radiation (with a wavelength of l = 1,54,056 Å) with a power of up to 2 kW (kilowatts)
M – 1 copper	Class: Copper. The M1 copper alloy contains a minimum amount of impurities (0,01%)

Introduction

Static hydrogen fatigue (SHF), namely, the reduction of the long-term strength of steel as a result of hydrogen embrittlement under statically loaded metal, representing one of the main types of corrosion damage to BCC iron-based alloys [1–3], is well known. The appearance of SHF in the metallurgical production method is explained by the capture of hydrogen gas from the atmosphere of areas of microscopic pores that occur near extended structural defects of metals of the iron subgroup, such as dislocations and grain boundaries, after the crystallization process in castings is completed [3,4].

In accordance with modern physical and mechanical concepts of the nature of static metal fatigue, for delayed fracture due to hydrogen embitterment, which determines SHF, a combination of mechanical stresses and critical values of hydrogen concentration is required, at which the bond in the crystal lattice is sharply weakened [1,5,6].

One of the possible ways to combat the brittleness of metals is to trap hydrogen into traps [7], which can be vacancies, impurity atoms, dislocations, grain boundaries, pores, and many other structural defects [4,8]. Modern Metallurgy direction, providing hydrogen seizure control is the use of substitution impurities. This direction, called the hydrogen treatment of materials, is currently developing intensively [9].

The interaction of hydrogen with impurities in metals can occur for several reasons: a) elastic distortion of the lattice caused by the impurity dissolved in it; b) the difference between the interactions of hydrogen with matrix and impurity atoms.

The physic mechanical properties of electrolytic alloys differ significantly from the properties of the pure metals contained in them, as well as metallurgical alloys, this is due to their structure.

The physical and mechanical properties of electrolytic alloys differ significantly from the properties of the pure metals contained in them, as well as metallurgical alloys, this is due to their structure. For a number of alloys, their phase structure corresponds to the state diagram of metallurgical alloys.

However, electrolytic alloys are characterized by the formation of supersaturated solid solutions based on the more electropositive component, as well as changes in the boundaries of the existence of individual intermediate phases or their absence. Therefore, electrolytic alloys are usually in an unstable thermodynamic state, their phase structure and properties change after heating, compared with metallurgical alloys. Thus, a large number of defects in the structure of electrolytic alloys lead to increased microhardness. During the formation of solid solutions and intermetallic compounds, which is characteristic for metallic systems, their microhardness and electrical resistivity increases [6,10]. Another feature of the structure of electrolytic coatings is their large heterogeneity at the macro and micro levels [4,6,7,11]. These in homogeneities are represented as structural defects, which mainly interact with hydrogen and metal.

A considerable amount of publications is devoted to the study of the interaction of hydrogen with defects and impurities in metals and alloys, but most of the information accumulated so far obtained from the cited publications is based on studies of the physicochemical properties of metals and alloys obtained by the metallurgical method.

The purposes of this work are

- study of the electrochemical processes for producing Ni–B and Ni–In composites;
- study of the effect of boron and indium dopants on the hydrogen permeability of nickel-based complexes (Ni_x-B_y-H_z and Ni_x-In_y-H_z), synthesized by the electrochemical method.

The main task is to show that by creating traps (structural, impurity) for hydrogen, for example, by additionally introducing chemical elements into the base metal structure or by changing the structure (intermetallic compounds), as well as using other methods, one can change the hydrogen solubility of the metal up or down, depending from technical application tasks.

Process of electrochemical formation of hydride compounds

The interaction of metals with hydrogen is distinguished by a number of characteristic signs during of their cathode reduction. First of all, this applies to those metals from the system of elements D.I. Mendeleev, who are most prone to the formation of metal-hydrogen bonds. These include the subgroups IVB, VB, VIB, VIIIB, as well as platinum-group elements specifically interacting with hydrogen and some rare-earth metals. The probability of formation of hydrogen compounds at the cathode depends on a parameter such as the overvoltage of the hydrogen produced. The amount of overvoltage on the electrode depends on the nature of the electrode, the current density, the composition of the solution and other factors. The magnitude of the overvoltage is different for different electrochemical processes.

A huge contribution to the theory of extraction processes and on electrodes from various materials was made by scientists under the guidance of A.N. Frumkin [5]. The authors showed that mobility, and, consequently, the electrical conductivity of ions and anomalously large. Due to the small size of the proton and the large electric field around it, a covalent bond arises between the proton and the indivisible electron pair of oxygen electrons of the molecule according to the scheme [5,12]:

$$H^+ + H_2O \rightarrow H_3O^+$$

As a result of this transformation, a hydroxonium cation H_3O^+ is formed. By the method of spectral analysis, it has been experimentally established that the reaction of formation H_3O^+ is exothermic, while the heat of reaction takes on a relatively high value:

$$H^+ + H_2O \rightarrow H_3O^+ + 1110\ kJ$$

The cathodes recovery H_2 and anodic oxidation of molecules H_2O and anions OH^-has a significant effect on the process of gas evolution on electrodes. Therefore, the task of increasing the intensity of gas excretion is reduced to a decrease in overvoltage's of H_2andO_2, released on the electrodes ($\eta_{H_2}$and η_{O_2}) [6]. The composition of the material of the electrodes affects the reduction and oxidation processes H_2O and OH^- ions. The value of η_{H_2} is related to the current density (i) by the well-known expression [5,12]:

$$i = k \cdot \exp\left(\eta_{H_2}/b\right) \quad (1)$$

where, a, b, k - are constants, while $k = \exp\left(-a/b\right)$, i - is the current density, A/cm^2.

The constant a - is a characteristic of the degree of irreversibility of the process at the electrode, which depends essentially on the material of the electrode and is equal to the magnitude of the overvoltage at $i = 1$ A/cm^2.

The value of b depends little on the electrode material and is a characteristic of the electrochemical process only. The relationship (2) between the constant a and the compressibility factor χ of the metal has been established [5,13], (Fig. 1):

$$a = 2 - \frac{1}{\sqrt{\chi \cdot 10^{-6}}} \quad (2)$$

Analysis of this dependence makes it possible to make the optimal choice of electrode material from the standpoint of hydrogen overvoltage. Depending on the minimum interatomic distance r when comparing the values of η_{H_2}, it was established [5,11–13] that the distance 2,7 Å corresponds to the extreme minimum of the value of η_{H_2}. As a rule, the

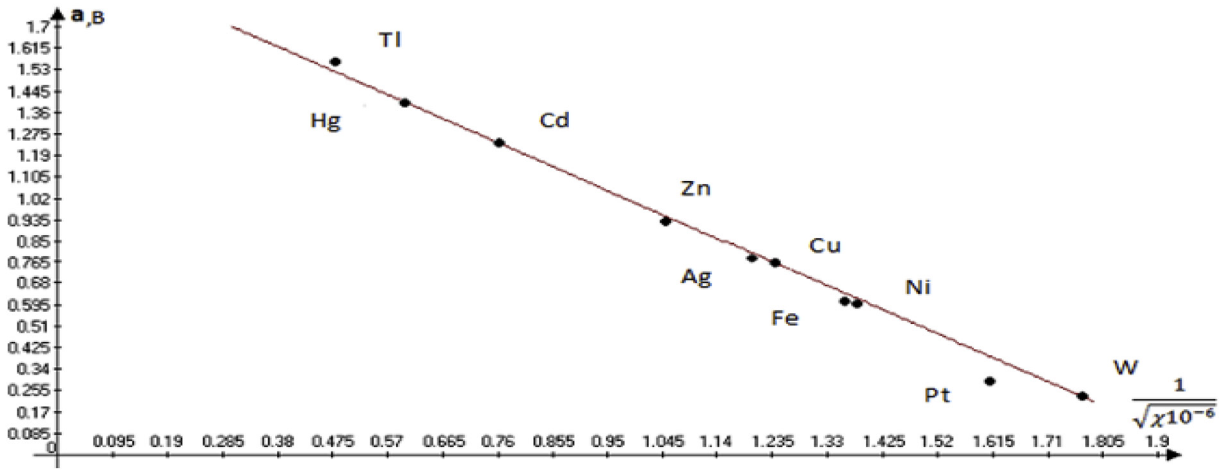

Fig. 1 – Dependence of the overvoltage of hydrogen allotment (η_{H_2}) on the compressibility factor χ of various metals at a current density of i = 1 A/cm².

temperature of the electrode differs significantly from the temperature of the electrolyte in volume due to various thermal effects therefore; the effect of temperature on the value of η_{H_2} must be taken into account. This effect can be quantified [5,12] using expression (3):

$$\left(\frac{\eta_{H_2}}{t}\right)_i = \frac{da}{dt} + \frac{R}{\lambda F} \ln i \quad (3)$$

where: F = 96,485 Kl - Faraday constant; λ - is the fraction of the current carried by one of the ions (in the absence of concentration changes), is the transport property of the solution, known as the transfer number; 8314 J/mol•K - the universal gas constant value.

Since $da/dt < 0$, the temperature overvoltage coefficient in its absolute value becomes more at low current densities, but less at high ones.

Among the features of hydrogen evolution at the cathode, one should add the presence of heat-generation effects, which are a consequence of chemical and electrochemical reactions occurring at the cathode [11].

On this basis, it can be assumed that during the dehydration of the hydroxonium cation, the absorbed energy will create a negative temperature gradient in the reaction zone, facilitating the transport of ions to the reaction zone. However, contrary to expectations, this phenomenon is not observed, due to the fact that in the near-electrode zone the most likely process is the disintegration of hydrogen atoms with the formation of hydrogen molecules from atoms. This reaction is accompanied by the release of heat, so it partially compensates for the absorption of thermal energy as a result of the dehydration of the hydroxonium cation. Another compensating component can be the energy of chemisorption of ions when they interact with the surface of metals, which likewise causes thermal diffusion of hydrogen cations into the depth of the electrolyte solution.

Analyzing the dependence of the overvoltage on the nature of the metal, it is easy to establish that Ni, having an insignificant overvoltage in comparison with other metals, for example, zinc (Fig. 1), can interact with atomic hydrogen and form hydride compounds. In addition, hydrogen protons, having small sizes (~10^{-5}Å) when penetrating into the crystal lattice, can diffuse into it over relatively large distances, which increase the probability of formation of hydride compounds in the bulk of the metal [14,15].

The metals of the iron subgroup (iron, cobalt and nickel) have an overvoltage that reaches noticeable values even at low current density. For example, overvoltage of nickel in a $NiSO_4$ solution at a current density of 0,1 mA/cm² is 0,3 V [6,16], and in a nickel sulphamate solution at i_k = 2 A/dm² ~ 0,4 V [10,17]. Compared with other metals of the periodic system of Mendeleev D.I., metals Cr, Fe, Co, Ni, Mn, Re and a number of others, characterized by a higher cathode overvoltage (more than 0.1 V), ююcan have a fine-crystalline structure, characterized by a greater length of intergranular boundaries and, accordingly, have a greater number of structural defects.

Methodical and instrumental support of experimental researches

Nickel-based alloys were selected as the object of study with catalytic activity and a tendency to hydrogen occlusion. The studies were carried out according to traditional methods [8,10,17] on samples in the form of films produced by electroplating.

To fabricate a Ni–B sample, the nickel plating sulphamate electrolyte was used in the form of boron compound of the class of higher polyhedral borates - $Na_2B_{10}H_{10}$.

Electrolysis modes: current density (i_k) 0,5–4,0 A/dm²; electrolyte temperature ($t_{el\text{-}ta}$) 30–50 °C; pH – 3,5–4,5. Anodes - nickel. Cathodes - copper grade M-1. The boron content in the Ni–B composite was determined by the spectrophotometer method [18].

For a standard electrolyte, a mode with the following parameters was chosen: the cathode current density is 2 A/dm², the acidity is 4,0 pH units, the electrolyte temperature is 40 °C, and the boron-containing additive concentration in the

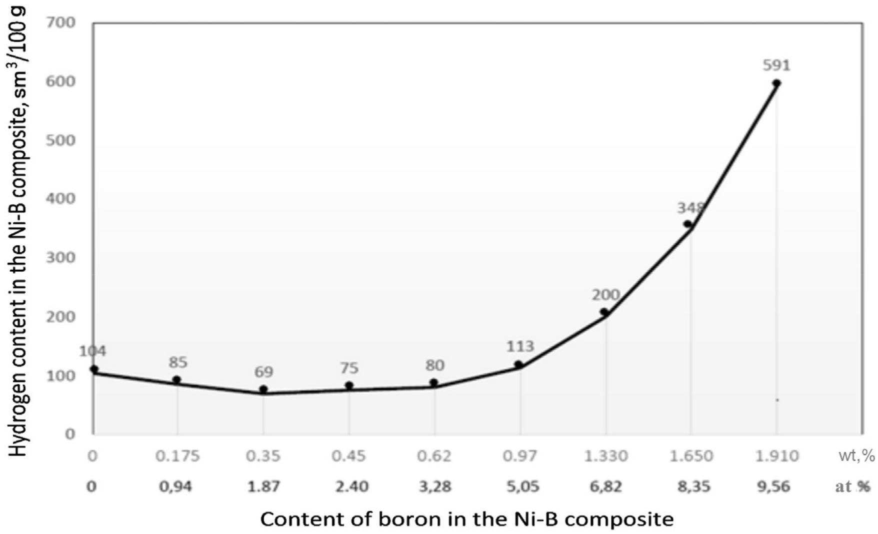

Fig. 2 – Dependence of hydrogen content in electrochemical composite Ni–B–H on boron concentration. Modes of electrolysis: $i_k = 2$ A/dm^2; $t_{эл\text{-}та} = 40$ °C; pH = 4,0; d = 4 μm; base–copper grade M-1.

electrolyte is 0,1–2 g/L. The percentage of boron in the sample was (from 0,1 to 2) wt.%.

The amount of hydrogen in Ni–B crystals was determined by the vacuum extraction method: the sample was placed in a quartz chamber containing a nitrogen «trap» in which a vacuum of 10^{-5} mm Hg was created, then the sample was heated up to 500 °C temperature. The volume of extracted hydrogen was calculated from the difference between its final and initial pressure in the measuring system using the ideal gas equation of state.

This installation provides an initial vacuum of at least 10^{-6} mm. Hg, the change in temperature of extraction from 20 °C to 800 °C. The relative error in determining the volume of hydrogen in the sample did not exceed 5%, while the substrate itself introduced a slight error, since copper adsorbs little the hydrogen [1–3].

Ni–In composites were made in a similar way - by electrolytic deposition on copper substrates (0,05 mm thick).

Electrolyte composition: $NiSO_4 \times 7H_2O = 140$ g/L; $Na_2SO_4 \times 10H_2O = 20$ g/L; $In_2(SO_4)_3$, the concentration of which was varied from 1 to 12 g/L, plus composite additives [19]. Electrolysis was carried out with a platinum anode and a copper cathode. The variation in the amount of $In_2(SO_4)_3$ provided the preparation of Ni–In composites with different ratio of components.

The content of components in the Ni–In system was determined by the X-ray fluorescence method. X-ray phase analysis was performed using a diffractometer DRON-2,0.

In order to eliminate the influence of the so-called background hydrogen in the samples and in the chamber of targets on the accuracy of the results obtained, the experiments used a hydrogen isotope, deuterium.

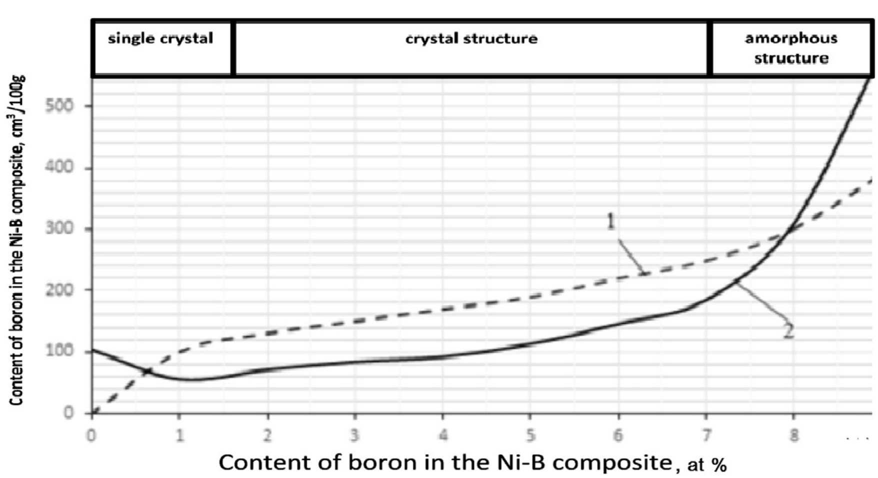

Fig. 3 – Regularity of change in the ability to hydrogen occlusion for electrochemical systems (metals and alloys): 1- a hypothetical metal; 2 - electrochemical composite Ni–B–H.

Table 1 – Results of the calculation of radiographs.

No	2Θ	Θ	sinΘ	d/n	Phase	Intensity
1	2	3	4	5	6	7
Sample number 1 (In = 0 g/L)						
1	39,21	19,60	0,3355	2,2950	Ni	Very weak
2	45,47	22,73	0,3864	1,9927	Ni	Weak
3	66,00	33,00	0,5446	1,4138	Ni	Very strong
4	74,53	37,26	0,6055	1,2716	Ni	Very weak
5	79,75	39,87	0,6411	1,2011	Ni	Weak
Sample number 2 ($In_2(SO_4)_3$ = 0,5 g/L)						
1	39,37	19,68	0,3368	2,2862	Ni	Very weak
2	45,46	22,73	0,3864	1,9927	Ni	Very weak
3	66,18	33,09	0,5459	1,4105	Ni	Very strong
4	74,37	37,18	0,6044	1,2739	Ni	Weak
5	79,65	39,82	0,6404	1,2023	Ni	Weak
Sample number 3 ($In_2(SO_4)_3$ = 1 g/L)						
1	37,5	18,75	0,3214	2,3955	Ni	Very weak
2	45,42	22,71	0,3861	1,9945	Ni	Very weak
3	66,38	33,19	0,5474	1,4066	Ni	Very strong
4	74,33	37,16	0,6040	1,2748	Ni	Very weak
5	79,67	39,84	0,6406	1,2019	Ni	Weak
Sample number 4 ($In_2(SO_4)_3$ = 2 g/L						
1	37,44	18,72	0,3209	2,3992	γ-$InNi_3$	Weak
2	45,42	22,71	0,3861	1,9945	Ni	Weak
3	53,86	26,93	0,4529	1,7002	$InNi_2$	Weak
4	66,00	33,00	0,5446	1,4138	Ni	Very strong
5	67,36	33,68	0,5546	1,3885	η-$In_{27}Ni_{10}$	Weak
6	79,67	39,84	0,6406	1,2019	Ni	Weak
Sample number 5 ($In_2(SO_4)_3$ = 4 g/L)						
1	30,40	15,20	0,2622	2,9369	η-$In_{27}Ni_{10}$	Average
2	32,90	16,45	0,2832	2,7192	In	Average
3	38,88	19,44	0,3328	2,3137	γ-$InNi_3$	Average
4	39,16	19,58	0,3351	2,2978	In	Average
5	43,30	21,65	0,3689	2,0873	In_3Ni_2	Weak
6	45,42	22,71	0,3861	1,9945	Ni	Weak
7	53,75	26,87	0,4519	1,7037	$InNi_2$	Weak
8	66,38	33,19	0,5474	1,4066	Ni	Very strong
9	67,36	33,68	0,5545	1,3886	η-$In_{27}Ni_{10}$	Weak
10	79,86	39,93	0,6418	1,1998	Ni	Average
Sample number 6 ($In_2(SO_4)_3$ = 8 g/L)						
1	27,65	13,82	0,239	3,2217	ε –InNi	Strong
2	30,00	15	0,2588	2,9752	$InNi_2$	Very strong
3	30,62	15,31	0,264	2,9166	η-$In_{27}Ni_{10}$	Average
4	33,12	16,56	0,285	2,7010	In	Strong
5	35,46	17,73	0,3045	2,5287	In_3Ni_2	Weak
6	39,75	19,87	0,3399	2,2653	Ni	Average
The rest of the table						
7	43,75	21,87	0,3725	2,0671	η-$In_{27}Ni_{10}$	Weak
8	50,93	25,46	0,43	1,7906	δ'-In_3Ni_2	Average
9	52,50	26,25	0,4422	1,7412	γ-$InNi_3$	Average
10	60,93	30,46	0,507	1,5187	ε –InNi	Weak
11	66,25	33,12	0,5464	1,4090	Ni	Very strong
12	67,75	33,87	0,5573	1,3810	ε –InNi	Average
13	75,78	37,89	0,6141	1,2538	η-$In_{27}Ni_{10}$	Average
14	79,37	39,68	0,6384	1,2060	η-$In_{27}Ni_{10}$	Average
15	79,68	39,84	0,6406	1,2010	Ni	Weak
Sample number 12 ($In_2(SO_4)_3$ = 12 g/L)						
1	27,81	13,90	0,2403	3,2040	ε-InNi	Strong
2	30,00	15,00	0,2588	2,9750	$InNi_2$	Very strong
3	30,62	15,31	0,2640	2,9160	η-$In_{27}Ni_{10}$	Average
4	32,96	16,48	0,2837	2,7141	In	Strong
5	35,34	17,67	0,3035	2,5370	In_3Ni_2	Very weak
6	37,65	18,82	0,3227	2,3860	In_3Ni_2	Strong
7	39,78	19,89	0,3402	2,2633	Ni	Weak
8	43,75	21,87	0,3725	2,0671	η-$In_{27}Ni_{10}$	Weak
9	50,78	25,39	0,4287	1,7961	Δ-In_3Ni_2	Weak

Table 1 – (*continued*)

No	2Θ	Θ	sinΘ	d/n	Phase	Intensity
10	52,65	26,32	0,4435	1,7360	γ-$InNi_3$	Weak
11	54,12	27,06	0,4549	1,6926	In	Weak
12	56,71	28,35	0,4749	1,6210	Ni_3In	Very weak
13	57,34	28,67	0,4797	1,6050	δ'-In_3Ni_2	Very weak
14	58,90	29,45	0,4916	1,5663	γ-$InNi_3$	Very weak
15	60,00	30,00	0,5000	1,5400	γ-$InNi_3$	Very weak
16	61,87	30,93	0,5140	1,4980	η-$In_{27}Ni_{10}$	Weak
17	66,09	33,04	0,5452	1,4123	Ni	Very weak
18	67,65	33,82	0,5566	1,3833	ε –InNi	Weak
19	70,46	35,23	0,5769	1,3347	γ-$InNi_3$	Very weak
20	73,43	36,71	0,5978	1,2880	$InNi_2$	Very weak
21	75,93	37,96	0,6152	1,2516	δ'–In_3Ni_2	Very weak

The kinetics of the development of the deuterium desorption spectrum from samples of the Ni–In system, depending on the ratio of components and dose of implanted deuterium, was studied. The irradiation of the samples and measurements of the spectra of thermal desorption were carried out on the experimental setup «SKIF», described in detail in Refs. [19]. The samples were mounted on a tantalum foil heater of 5 × 45 × 0,1 mm^3. The temperature was measured using a tungsten rhenium thermocouple BP5/20 attached to a heater. The introduction of deuterium into the samples was carried out by implantation of deuterium ions with energy of 12 keV with radiation doses within 3×10^{17}–3×10^{18} at. D/cm^2.

Experimental results and discussion

Formation and study of the Ni–B composite structure.

For efficient accumulation of hydrogen in the volume of structural defects, the foil (tape) made of Ni_x-B_y-H_z composites is most often used, in which the content of dissolved hydrogen is increased by changing the content of the boron doping element [8,10,20].

It should be noted that the probability distribution of the density of defects in electrochemical systems depends on the parameters of electrolysis. In this case, the chemical composition of the electrolyte (the presence of an amorphytization agent - boron) and the cathode current density are the determining factors. This section presents the results of a study of the effect of boron concentration on the hydrogen content in the nickel-boron-hydrogen system.

It has been established (Fig. 2) that when small amounts of boron are added to Ni up to 2,1 at. %, the volume of hydrogen in the metal decreases and Ni becomes more ductile, and the resulting alloy has better physical and mechanical properties (solderability when using acid-free fluxes, weldability with Al conductors, low transient resistance) [10].

When increasing the amount of boron more than 4 at.%, the hydrogen content increases, thus there is a change in the structure and a number of physicochemical properties. Nickel coating with such a quantity of boron increases the hardness, wear resistance, heat resistance of objects. This phenomenon is associated with the peculiarities of cathode

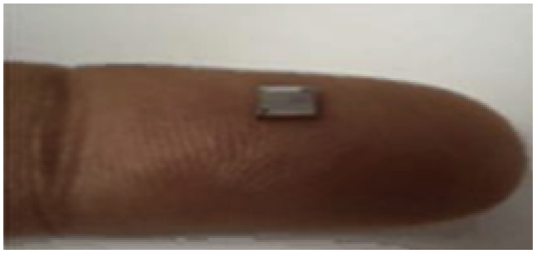

Fig. 4 – The visual appearance of the sample saturated by hydrogen by electrochemical method.

electroreduction of nickel in the presence of boron-containing reducing agents and - with structural changes.

At a relatively low concentration, boron, penetrating into the FCC nickel lattice, concentrates along the boundaries of the crystallites and the nickel lattice defects, blocks them, and this hinders the inclusion of hydrogen. When the boron concentration is more than 7 at.%, A transition from the crystalline structure to the amorphous is observed, additional defects appear in the FCC nickel lattice, which are the basis for the incorporation of hydrogen, reaching a value of about 600 cm^3/100 g when the boron content is 9 at.%. The increase in the content of hydrogen in Ni–B is associated with the capture of hydrogen around boron and their greater interaction than with nickel, and, as a result, with the subsequent formation of complexes. Therefore, it is obvious that boron is an impurity trap for hydrogen atoms.

The relationship of the structural changes occurring in nickel due to the introduced boron with the amount of hydrogen included is of interest, therefore, the influence of the structure on the occluding ability of the metal was studied. This ability is evaluated in terms of changes in the number of defects in the metal structure [21,22]. For better comparison Fig. 3 separately depicts curve 1 of a certain hypothetical alloy with increasing defects and zones of the structural state.

For a hypothetical metal, when passing from a pure single crystal to an amorphous structure, the degree of imperfection of the structure increases and the probability of hydrogen absorption increases (Fig. 3, curve 1). In one of the first works on this topic [22], it was proposed to evaluate the ability of metals and alloys to occlude hydrogen with structural changes and to separate the process of metal absorption by hydrogen into 3 phases of state: single crystal, crystalline and amorphous. The increase in the hydrogen content in the amorphous state of the metal, the authors attributed to an increase in the degree of imperfection of the structure. Experimental confirmation of the hypothesis of [22], were the results also presented in (Fig. 3, curve 2).

The pattern of changes in the ability of the Ni–B–H electrochemical composites to hydrogen occlusion is reflected in Fig. 3 (curve 2). With the introduction of boron in nickel from 2 at. % to 10 at. % there is a transition from the crystal structure (up to 5 at.%) to the implicitly expressed crystal structure (6–9) at. % with the continuation of the transition to amorphous one, which is accompanied by an increase in the extracted hydrogen from the sample, the content of which was determined by vacuum extraction. In the process of formation of nickel-boron composites with a fine-crystalline structure, in them the rate of nucleation exceeds the rate of their growth unlike pure nickel [5,6,20]. Synthesis of composites that form a combination of elements in the form of small crystallites, due to the appearance of structural and impurity traps in them creates conditions for the formation of internal structures with a markedly increased number of defects per unit volume.

The presence of boron in the alloy provides increased strength of the material [10], increases the number of defects that are «traps» for hydrogen atoms. Such systems are characterized by a small crystallite size, and their characteristic feature is the presence of a significant fraction of grain boundaries with their structural imperfections, which serve as the operational zone of the hydride formation process. This process develops initially on the electrode surface, and then, penetrating deeper, continues to spread in its bulk structure, thereby creating conditions for hydrogen accumulation [21,23–25].

It has been experimentally confirmed [8,10,20] that Ni–B–H composites are most resistant to thermal cycling, which is an important characteristic of efficiency in the case of using an alloy as a hydrogen storage ring. Samples of electrochemical composites were synthesized (Fig. 4), with a layer thickness of 8 μm and an area of 0,12 dm^2, a geometric form is a ribbon.

Extraction of hydrogen takes place within the range of temperature changes from 50 to 80 to 450 °C, at various choices of combinations of elements on chemical composition. The range of changes in the concentration of atomic boron in nickel is from 5 to 10 at.%. In this range of boron concentration, the monolithic state of the sample is preserved, without transition to the powder state.

An analysis of literature published over the past 15 years confirms the almost complete lack of information about the functional properties of nickel-indium electrochemical composites. Unfortunately, the available information [19,26–28]

Table 2 – The dependence of the phase composition of nickel - indium electrochemical composites on indium content.

Sample number	Content $In_2(SO_4)_3$ in electrolyte, g/L	The indium content in the composite Ni–In wt.%	Ni	$InNi_2$	$InNi_3$	In_3Ni_2	$In_{27}Ni_{10}$	In	InNi
1	2	24	+	+	+	+	+	−	−
2	4	38	+	+	+	+	+	+	−
3	8	45,6	+	+	+	+	+	+	+
4	12	61,3	−	+	+	+	+	+	+

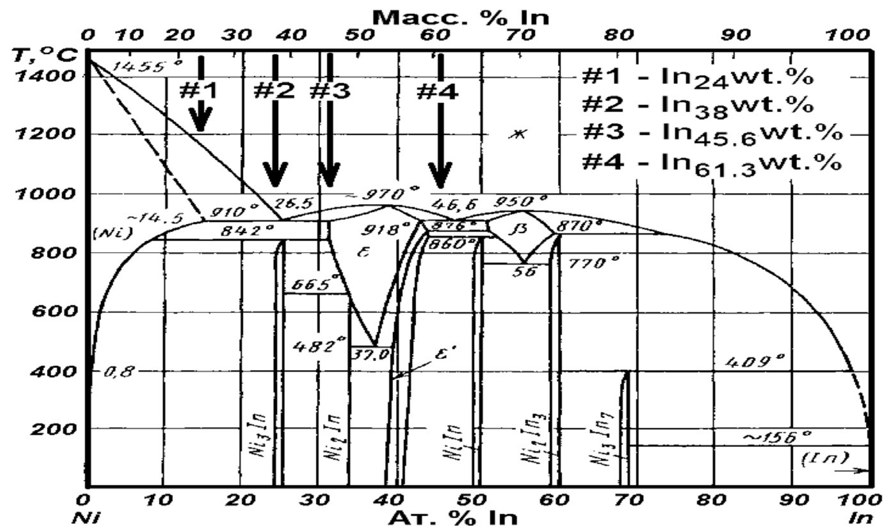

Fig. 5 – The diagram of the phase state of the Ni–In system and experimental samples of Ni–In composites: No 1–24 wt% In; No 2–38 wt% I n; No 3–45,6 wt% In; No 4–61,3 wt% In.

does not confirm the ability of the Ni–In alloy to form the metal hydride phase and targeted accumulation of hydrogen.

The interaction of hydrogen with intermetallic compounds was described in Ref. [29], and it was shown that if a new phase is incoherently associated with the Ni matrix through a network of dislocations, that this system can be considered as a structural trap for hydrogen atoms.

Such a structure in the form of an intermetallic compound of nickel with indium must first be formed, and then the ability of the formed structure to occlude hydrogen be evaluated.

Based on the data obtained from the studied structures of Ni–In formed electrochemical composites, the electrolyte composition was initially determined to solve this problem.

The optimum deposition mode for nickel coatings was experimentally selected: the cathode current density is 2 A/dm^2, the electrolyte temperature is room temperature. To determine the effect of each of the parameters of electrodeposition on the properties of the samples, only one parameter was varied: either the cathodic current density or the indium concentration in the electrolyte.

Samples of the following concentrations were made: No. 1–24 wt% In (at a concentration of $In_2(SO_4)_3$ 1 g/L); No. 2–38 wt % In (at a concentration of $In_2(SO_4)_3$ 2 g/L); No. 3–45,6 wt% In (at a concentration of $In_2(SO_4)_3$ 4 g/L); No. 4–61,3 wt% In (at a concentration of $In_2(SO_4)_3$ 12 g/l). In the phase state diagram (Fig. 5), the arrows mark the samples that were synthesized and studied [19].

The results of x-ray analysis are presented in Table 1.

With an increase in the concentration of $In_2(SO_4)_3$ in the electrolyte from 1 to 12 g/L, the content of In the composite increases to 61,3 wt%, and the Ni content, respectively, becomes equal to 38,7 wt%

When the content in the electrolyte $In_2(SO_4)_3$ in an amount of less than 2 g/L, nickel is deposited on the cathode, the presence of pure indium and its compounds with nickel is not marked. By increasing the content of indium (III) sulfate in the solution to 2 g/L, at the cathode we obtain the deposition of pure nickel and intermetallic compounds: $InNi_2$, $InNi_3$, In_3Ni_2, η-$In_{27}Ni_{10}$ [28,30,31].

In Ni–In electrochemical composites obtained from electrolytes containing $In_2(SO_4)_3$–8 g/L, an additional ε-InNi phase appears.

With an increase in the indium (III) sulfate content in the electrolyte up to 12 g/L in the formed Ni–In composite, the line corresponding to pure nickel is absent. When interpreting the X-ray diffraction patterns, the phase corresponding to the η-$In_{27}Ni_{10}$ intermetallic compound was detected; however, in the diagrams of the equilibrium state, according to Ref. [31], this phase is not shown. Possibly, this is a non-equilibrium phase.

The electrochemical system was created at room temperature; therefore, the absence of high temperatures in the manufacture of the Ni–In composite obviously contributed to the stabilization of the η-$In_{27}Ni_{10}$ phase. Presumably, the η-phase is formed by any synthesis methods that exclude heat.

The summarized results of the study are presented in Table 2 for four samples in which new phases were discovered - intermetallic compounds.

An increase in the indium concentration in composites (more than 30 wt% In) is accompanied by the appearance and increase in the intensity of additional lines in diffraction patterns. This is due to the formation of Ni_3In and In_2Ni intermetallic compounds (the possibility of their formation is indicated by the phase diagram of the Ni–In system) [19,28,30,31].

For further studies, the concentration of indium sulfate in the electrolyte was 4 g/L; at this concentration, samples with indium content of 38 wt% and nickel 62 wt%. In terms of atomic percentages, it corresponds to $Ni_{70}In_{30}$ stoichiometry.

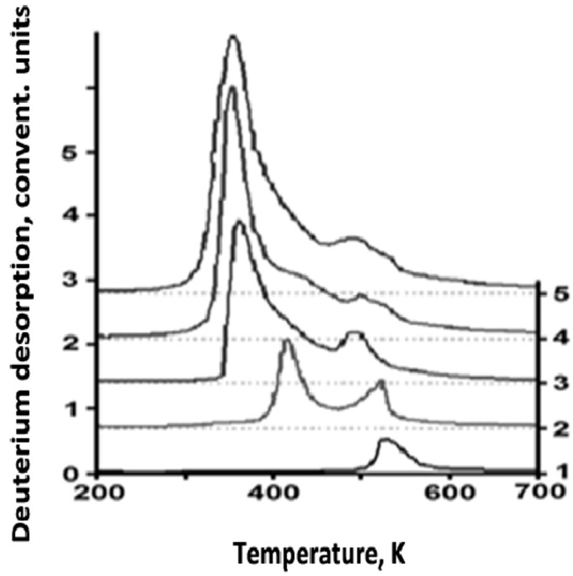

Fig. 6 – The thermal desorption spectra of deuterium implanted in composite samples $Ni_{70}In_{30}$: 1–3 × 10^{17} D/cm^2; 2–7,5 × 10^{17} D/cm^2; 3–1,3 × 10^{18} D/cm^2; 4–2 × 10^{18} D/cm^2; 5–3 × 10^{18} D/cm.^{2}.

Study of thermal desorption of hydrogen from composites Ni_{70}–In_{30}-H_x

In Fig. 6 the spectrum of thermal desorption of deuterium implanted in $Ni_{70}In_{30}$ composite samples is presented.

On the spectrum, one can distinguish the temperature ranges of desorption depending on the dose of implanted deuterium in composite materials. The type of spectrum of TDS of deuterium depends on the implantation dose. At low doses of implanted deuterium, there is one peak in the spectrum with a maximum temperature of ~530 K. A dose increase leads to the appearance of a new low-temperature peak with a maximum temperature of ~420 K. An increase in the concentration of deuterium leads to the formation of a solid solution of deuterium in the $Ni_{70}In_{30}$ composite, the decomposition temperature of which in vacuum is ~530 K, the decomposition temperature of another deuterium-containing phase is ~350 K.

The deuterium content for a composite with $Ni_{70}In_{30}$ phase composition is 2 at. D/at. Meth., which corresponds to 5,3 wt % D [19].

Conclusion

A technology has been developed that, using an electrochemical method, ensures the formation of the structure of a nickel alloy with a high degree of imperfection, with the use of boron additives, allowing obtaining a fine-crystalline structure of the alloy. The process of formation of the $Ni_xB_yH_z$ composite by an electrochemical method is shown, in which, with an increase in the boron content, the solubility of the incorporated hydrogen improves. The studies were carried out in the range of changes in concentrations and atomic boron in nickel from 5 at. % to 10 at. %, while the hydrogen content in the alloys was about 600 cm^3/100 g, while maintaining the monolithic structure of the sample (without a transition to the powder state). To measure the concentration of hydrogen in the matrix, a vacuum extraction method was used.

The phase composition of Ni_x-In_y electrochemical composites is investigated. As a result, the electrolyte composition was selected with the optimal content of indium sulfate in the electrolyte, which made it possible to produce fine-crystalline and equi-grained $Ni_{70}In_{30}$ composites that are capable of retaining implanted deuterium. The optimal mode of deposition of nickel-indium composites was experimentally selected: cathode current density, electrolyte temperature.

The spectra of deuterium thermal desorption from $Ni_{70}In_{30}$-Dx composites were obtained, which made it possible to separate the temperature ranges of desorption of ion-implanted deuterium depending on the implantation dose. As a result, the possibility of obtaining samples with deuterium content up to 2 at. D/at. Meth, which corresponds to 5,3 wt % (for composites of this compound).

In conclusion, it should be noted that an increase in the content of additives of boron or indium in nickel-based electrochemical composites creates conditions for the growth of structural defects in crystals, which leads to an increase in the content of hydrogen in them.

Thus, creating hydrogen traps in the metal, you can prepare a metal base to saturate it with hydrogen. Further studies of the hydrogen-accumulating properties of metals and alloys should include: studying the patterns of absorption, accumulation and thermally activated hydrogen evolution, thermodynamic parameters (temperature ranges of retention, activation energy of thermal desorption) depending on the nature and concentration of the nano-forming additive in the synthesized materials. The research results will be useful for an in-depth understanding of the structure formation processes of nanomaterials and the formation of a strategy for synthesizing new multifunctional nanomaterials.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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